8/24/26

Comparing 2 ISA models: Sol vs claude opus- sol was simpler and easier to read in the chatISA

8/26/26
Prescriptive analytics - have a model, with some error, i can predict how things will act in the future

Unit of analysis - what does a row of data mean

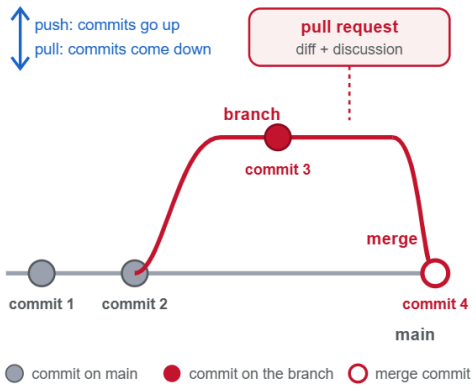
Encoding- how are you taking rows of data and representing them in the dataset (rows DOES NOT = crashes)

Visual encoding - how each variable in my dataset is represented (color, height, etc…)

Git - it is a tool, it runs on your machine and works offline. R files, Python files, you can revert and fix mistakes.

Github - its the place, copies live there so they can be sharedm reviewed and published (like the google drive for git)

repository (repo) - a folder Git watches
Commit - a named snapshot of that folder (like a google doc, can look at the history and see the edits, the commit saves the history or whats changed) put descriptions of what you've changed
push / pull - upload / download snapshots to or from GitHub
Branch- a parallel copy for work in progress
pull request - "please merge my branch," plus discussion
Merge - accept it into main



(https://aashbhar.github.io/portfolio/)
8/31/26
The analytics journey has 4 stages, business intelligence lives in **pre-analytics and descriptive**
Encoding - fancy way of saying "how does a variable from my file get represented in the chart"

- Color is not used to encode counties, “”not say this county is this colro adn this is this color”

Edit readme.md - when you saved the readme.md then click “commit to main” in github desktop, it is ONLY on the computer used

Git line change - the file that shows which line has changed in git, is similar to “class02.R”

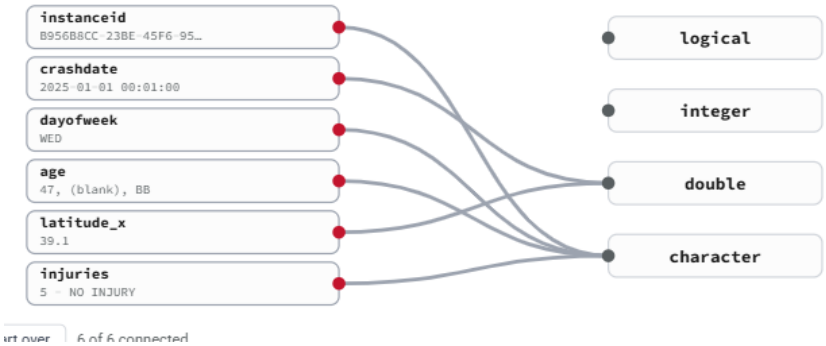
R basics-
YAML = yet another markdown language
Knit = goes straight into the folder (markdowns folder)
TOC = table of contents

6 types of atomic vectors
- Logical, integer, double and character are the most common
  - Logical: true or false
  - Integer: whole numbers
  - Double: any number
  - Character: text
- Complex and raw
  - Complex: imaginary numbers
  - Raw: bytes

instanceid	crashdate	dayofweek	age	latitude_x	injuries
46619FB1	2025-11-10 08:53:00	MON	47	39.138116	4 - POSSIBLE INJURY
0307C584	2025-09-08 15:50:00	MON	32	39.160639	5 - NO APPARENTY INJURY
09803D2E	2025-07-27 12:24:00	SUN	23	39.137162	5 - NO APPARENTY INJURY
A0583304	2025-10-24 20:40:00	FRI	32	39.128541	5 - NO APPARENTY INJURY
2DBB2FF7	2025-09-12 12:42:00	FRI	BB	39.08662	5 - NO APPARENTY INJURY
F9B159B9	2025-09-20 14:52:00	SAT	45	39.101339	5 - NO APPARENTY INJURY
7B5B1C68	2025-10-06 10:18:00	MON	27	39.196296	5 - NO APPARENTY INJURY

ortened to its first 8 characters to fit. NA means the value is missing. crashdate is understood as a date when using readr::read_csv().

ın on the left to a type on the right (or click one, then the other). Each column gets exactly one type. Then press Check.



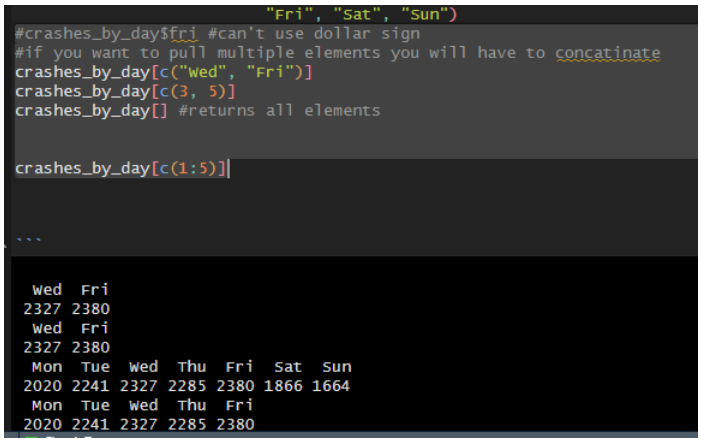
urt over 6 of 6 connected

Which file you should click on before opening rstudio in the class - **business_intelligence.Rproj**

Logical operator (crashes_by_day > 2000)

Atomic Vector - one dimensional data structure

- Each vector COULD be of a diff type (char, double, logical etc…)



```
                        "Fri", "Sat", "Sun")
#crashes_by_day$fri #can't use dollar sign
#if you want to pull multiple elements you will have to concatinate
crashes_by_day[c("Wed", "Fri")]
crashes_by_day[c(3, 5)]
crashes_by_day[] #returns all elements


crashes_by_day[c(1:5)]


```

```
 Wed  Fri
2327 2380
 Wed  Fri
2327 2380
 Mon  Tue  Wed  Thu  Fri  Sat  Sun
2020 2241 2327 2285 2380 1866 1664
 Mon  Tue  Wed  Thu  Fri
2020 2241 2327 2285 2380
```

crashes_by_day[c(1:5)]

Tibble vs **data frame**

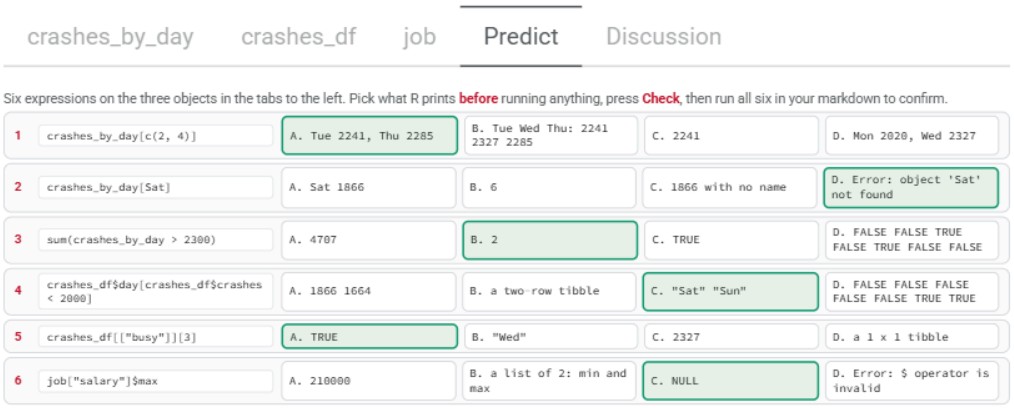
- DATA FRAME: Prints the entire dataset, which can flood your console if the data is large. Does not display data types.
    - Has rows and columns
- TIBBLE: Previews only the first 10 rows and columns that fit the screen. Explicitly lists data types

Glimpse - gives number of rows, columns, names (similar to str but prints it nicer)

List - flexible data structure

- Allows you to have things that aren't the same size
- One [ ] gives you the list of what youre interested in
- 2 [[ ]] gives you what you're visualizing

CHECK WORDING AND OPERATIONS to make sure they are written right



- Read_csv can read data that is locally stored or a web address
- Read_excel has to have the excel file on your computer
- fromJSON() reads a local .json file or one served at a web address

---

Class 5

Class 6

- Excel files are binary and R can read them
- fromJSON() on the cincy crash file returned a data frame where every column was character becasue the file **quotes every value, so R stores them as text**
- We are making another data folder because we want the data to come with the app, if they were in separate folders they wouldn't be packaged together

---

Class 7

The **pipe** takes the result of what's on its left and feeds it in as the first argument to the function on its right

App link:http://127.0.0.1:5632
Ctrl + shift+m = |>

Left_join = left table will keep all the rows, and combine common info and factors from right table to left table.

Inner_join = keep the intersectionbetween both tables, so only keep common factors, and add similar info to match em

Anti_join = shows what was lost

Assignment 5: https://midwest-airbnb-chat-i5uu.onrender.com/

Class 8

API = Application programming interface

- A way for you to communicate within code to programs and applications



Class 10
Scraping data

Website vs webpage
site(maimoh.edu)
page(html file)

SITE in domain part of code
Path to get to specific part in the path code

div p — every <p> inside a , at any depth
div > p — every <p> directly inside a (one level)

Class 10&11
Web scraping

Class 13 - Tidy and Technically correct
3 rules

- Each variable is a column
- Each observation is a row
- Each value is a cell

Tidy
1. Does each Location / Operator cell hold one value?
2. Does each Aircraft Type / Registration cell hold one value?
3. Does each Fatalities cell hold one value?
NO

Technically correct
4. Can we type every column name without quotes?
5. Is Date stored as a date?
6. Is Fatalities stored as a number?
NO